🚌 Project Data Engineer — Transjakarta Public Transportation Transaction

Dataset: Transjakarta Transportation Transaction (Kaggle)

Objective: Melakukan proses Data Engineering yang lengkap meliputi:

1. **Data Cleansing** & **Data Quality Check**
2. **Data Governance** & **Medallion Architecture** (Bronze → Silver → Gold)
3. **Privacy Policy** — Masking data sensitif
4. **Metadata Strategy** — Data Dictionary
5. **Analisis Data** & **Visualisasi** dengan insight
6. **Export** data bersih ke **Excel** dan **MySQL**

📖 Apa itu Data Governance?

Data Governance adalah kerangka kerja yang mencakup kebijakan, proses, standar, dan metrik yang memastikan pengelolaan data secara efektif dan efisien di seluruh organisasi.

Komponen utama Data Governance:

Komponen	Deskripsi
Data Quality	Memastikan data akurat, lengkap, konsisten, dan valid
Data Privacy	Melindungi data sensitif melalui masking, encryption, atau anonymization
Data Architecture	Menerapkan arsitektur data yang terstruktur (contoh: Medallion Architecture)
Metadata Management	Mendokumentasikan data melalui Data Dictionary dan data catalog
Data Lineage	Melacak asal-usul dan transformasi data dari sumber ke tujuan

Medallion Architecture

Medallion Architecture adalah pola arsitektur data yang membagi data menjadi **tiga layer**:

- 🥉 **Bronze Layer**: Data mentah (raw) langsung dari sumber tanpa transformasi
- 🥈 **Silver Layer**: Data yang sudah dibersihkan, divalidasi, dan di-enrichment
- 🥇 **Gold Layer**: Data yang sudah di-agregasi dan siap untuk analisis/reporting

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⚙️ Setup & Import Libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import hashlib
import warnings
from datetime import datetime

warnings.filterwarnings('ignore')
pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', 100)
pd.set_option('display.float_format', '{:.2f}'.format)

# Styling
plt.style.use('seaborn-v0_8-darkgrid')
sns.set_palette("husl")
plt.rcParams['figure.figsize'] = (12, 6)
plt.rcParams['font.size'] = 11

print("✅ Libraries imported successfully!")
```

```
✅ Libraries imported successfully!
```

📂 Load Dataset

In [2]:
```
# Load dataset dari file CSV
DATA_PATH = r'Transjakarta - Public Transportation Transaction/dfTransjakarta180kRo
df_raw = pd.read_csv(DATA_PATH)
print(f"Dataset loaded: {df_raw.shape[0]:,} baris × {df_raw.shape[1]} kolom")
df_raw.head()
```

```
Dataset loaded: 189,500 baris × 22 kolom
```

Out[2]:

	transID	payCardID	payCardBank	payCardName	payCardSex	payCar
0	VRPJ892P3M98RA	3561407960318444	dki	Dr. Janet Nashiruddin, M.Ak	M	
1	ZWCH834I6M26HS	347728053419394	emoney	Balamantri Rahayu	M	
2	YRLD835V6L82GO	377105453850671	emoney	Dian Mustofa	F	
3	ZZBX143N6N83HQ	4486493302356581	dki	Cut Janet Suryatmi, M.Ak	M	
4	EWEG491A2W45DR	30139379978125	bni	dr. Mulyanto Pudjiastuti, M.Ak	F	

🔍 Data Understanding

Tahap pertama dalam Data Engineering adalah **memahami data** yang akan diolah. Kita perlu mengetahui struktur, tipe data, distribusi, dan kualitas data sebelum melakukan transformasi apapun.

In [3]:

```
# Informasi umum dataset
print("=" * 60)
print("INFORMASI DATASET")
print("=" * 60)
print(f"Jumlah Baris    : {df_raw.shape[0]:,}")
print(f"Jumlah Kolom    : {df_raw.shape[1]}")
print(f"Memory Usage    : {df_raw.memory_usage(deep=True).sum() / 1024**2:.2f} MB")
print()
df_raw.info()
```

```
============================================================
INFORMASI DATASET
============================================================
Jumlah Baris    : 189,500
Jumlah Kolom    : 22
Memory Usage    : 146.41 MB

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 189500 entries, 0 to 189499
Data columns (total 22 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   transID           189500 non-null  object
 1   payCardID         189500 non-null  int64
 2   payCardBank       189500 non-null  object
 3   payCardName       189500 non-null  object
 4   payCardSex        189500 non-null  object
 5   payCardBirthDate  189500 non-null  int64
 6   corridorID        182520 non-null  object
 7   corridorName      175972 non-null  object
 8   direction         189500 non-null  float64
 9   tapInStops        182259 non-null  object
 10  tapInStopsName    189500 non-null  object
 11  tapInStopsLat     189500 non-null  float64
 12  tapInStopsLon     189500 non-null  float64
 13  stopStartSeq      189500 non-null  int64
 14  tapInTime         189500 non-null  object
 15  tapOutStops       177131 non-null  object
 16  tapOutStopsName   182780 non-null  object
 17  tapOutStopsLat    182780 non-null  float64
 18  tapOutStopsLon    182780 non-null  float64
 19  stopEndSeq        182780 non-null  float64
 20  tapOutTime        182780 non-null  object
 21  payAmount         185782 non-null  float64
dtypes: float64(7), int64(3), object(12)
memory usage: 31.8+ MB
```

In [4]:

```python
# Statistik deskriptif
print("STATISTIK DESKRIPTIF — Kolom Numerik")
print("=" * 60)
df_raw.describe()
```

```
STATISTIK DESKRIPTIF — Kolom Numerik
============================================================
```

Out[4]:

	payCardID	payCardBirthDate	direction	tapInStopsLat	tapInStopsLon
count	189500.00	189500.00	189500.00	189500.00	189500.00
mean	394166420948976512.00	1990.16	0.50	-6.21	106.84
std	1272012666738117376.00	13.45	0.50	0.06	0.06
min	60404484630.00	1946.00	0.00	-6.39	106.61
25%	180004429530433.00	1982.00	0.00	-6.25	106.80
50%	3519005268452064.00	1991.00	1.00	-6.21	106.84
75%	4664460720356463.00	2001.00	1.00	-6.17	106.88
max	4999813783000465408.00	2012.00	1.00	-6.09	107.02

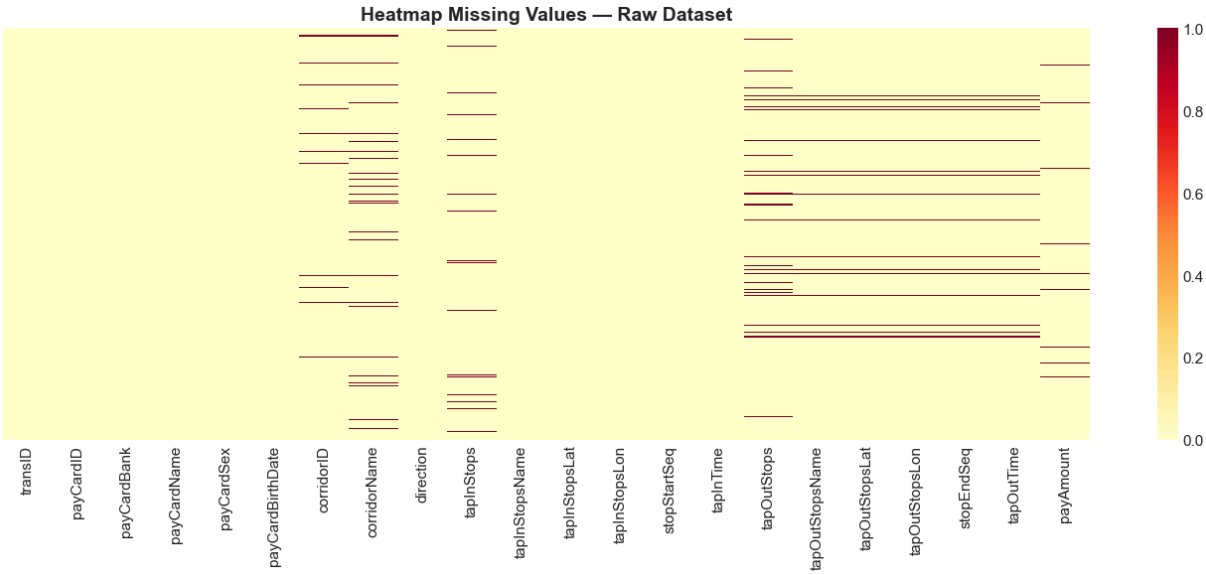
In [5]:

```
# Analisis Missing Values
print("ANALISIS MISSING VALUES")
print("=" * 60)
missing = df_raw.isnull().sum()
missing_pct = (df_raw.isnull().sum() / len(df_raw) * 100).round(2)
missing_df = pd.DataFrame({
    'Kolom': missing.index,
    'Jumlah Missing': missing.values,
    'Persentase (%)': missing_pct.values
}).sort_values('Jumlah Missing', ascending=False)
missing_df = missing_df[missing_df['Jumlah Missing'] > 0].reset_index(drop=True)
print(missing_df.to_string(index=False))
print(f"\nTotal kolom dengan missing values: {len(missing_df)}")
```

```
ANALISIS MISSING VALUES
============================================================
          Kolom  Jumlah Missing  Persentase (%)
   corridorName           13528            7.14
    tapOutStops           12369            6.53
     tapInStops            7241            3.82
     corridorID            6980            3.68
     tapOutTime            6720            3.55
     stopEndSeq            6720            3.55
 tapOutStopsLon            6720            3.55
 tapOutStopsLat            6720            3.55
tapOutStopsName            6720            3.55
      payAmount            3718            1.96

Total kolom dengan missing values: 10
```

In [6]:

```
# Visualisasi Missing Values — Heatmap
fig, ax = plt.subplots(figsize=(14, 6))
sns.heatmap(df_raw.isnull(), cbar=True, yticklabels=False, cmap='YlOrRd', ax=ax)
ax.set_title('Heatmap Missing Values — Raw Dataset', fontsize=14, fontweight='bold'
plt.tight_layout()
plt.show()
```



🥉 Bronze Layer — Raw Data Ingestion

Apa itu Bronze Layer?

Bronze Layer adalah **lapisan pertama** dalam Medallion Architecture. Data di layer ini adalah **data mentah (raw)** yang langsung diambil dari sumber tanpa modifikasi apapun.

Tujuan Bronze Layer:

- Menyimpan data asli sebagai *single source of truth*
- Menambahkan metadata ingestion (waktu masuk, sumber file)
- Menjadi titik awal untuk re-processing jika diperlukan

In [7]:
```
# === BRONZE LAYER ===
# Copy data mentah dan tambahkan metadata ingestion
df_bronze = df_raw.copy()

# Tambahkan metadata columns
df_bronze['_ingestion_timestamp'] = datetime.now().strftime('%Y-%m-%d %H:%M:%S')
df_bronze['_source_file'] = 'dfTransjakarta180kRows.csv'
df_bronze['_source_origin'] = 'Kaggle - Transjakarta Transportation Transaction'
df_bronze['_record_id'] = range(1, len(df_bronze) + 1)
df_bronze['_layer'] = 'bronze'

print(f"✅ Bronze Layer created: {df_bronze.shape[0]:,} rows × {df_bronze.shape[1]
print(f"   Metadata columns added: _ingestion_timestamp, _source_file, _source_orig
df_bronze.head(3)
```

```
✅ Bronze Layer created: 189,500 rows × 27 columns
   Metadata columns added: _ingestion_timestamp, _source_file, _source_origin, _reco
rd_id, _layer
```

Out[7]:

	transID	payCardID	payCardBank	payCardName	payCardSex	payCar
0	VRPJ892P3M98RA	3561407960318444	dki	Dr. Janet Nashiruddin, M.Ak	M	
1	ZWCH834I6M26HS	347728053419394	emoney	Balamantri Rahayu	M	
2	YRLD835V6L82GO	377105453850671	emoney	Dian Mustofa	F	

🥈 Silver Layer — Data Cleansing & Quality Check

Apa itu Data Quality Check?

Data Quality Check adalah proses sistematis untuk mengevaluasi dan memastikan kualitas data berdasarkan beberapa dimensi:

Dimensi	Deskripsi	Contoh Pengecekan
Completeness	Data tidak memiliki nilai kosong yang signifikan	Cek missing values pada setiap kolom
Consistency	Data seragam dan tidak bertentangan	Format tanggal yang sama, satuan yang konsisten
Validity	Data sesuai dengan aturan bisnis	Umur tidak negatif, koordinat dalam range valid
Uniqueness	Tidak ada duplikasi yang tidak diinginkan	Cek duplicate transID
Timeliness	Data masih relevan dan up-to-date	Cek range tanggal transaksi

Apa itu Data Cleansing?

Data Cleansing (pembersihan data) adalah proses mendeteksi dan memperbaiki (atau menghapus) data yang corrupt, tidak akurat, atau tidak relevan dari dataset.

In [8]:

```
# === SILVER LAYER — Step 1: Start from Bronze ===
df_silver = df_bronze.copy()
df_silver['_layer'] = 'silver'

# Hapus metadata columns yang tidak perlu untuk analisis
meta_cols = ['_ingestion_timestamp', '_source_file', '_source_origin', '_record_id'
```

```python
analysis_cols = [c for c in df_silver.columns if c not in meta_cols]

print("📊 DATA QUALITY CHECK — SEBELUM CLEANSING")
print("=" * 60)

# 1. Completeness Check
print("\n1️⃣ COMPLETENESS CHECK")
total_cells = df_silver[analysis_cols].shape[0] * df_silver[analysis_cols].shape[1]
missing_cells = df_silver[analysis_cols].isnull().sum().sum()
completeness = ((total_cells - missing_cells) / total_cells * 100)
print(f"   Total cells       : {total_cells:,}")
print(f"   Missing cells     : {missing_cells:,}")
print(f"   Completeness Rate : {completeness:.2f}%")

# 2. Uniqueness Check
print("\n2️⃣ UNIQUENESS CHECK")
duplicates = df_silver.duplicated(subset=['transID']).sum()
print(f"   Duplicate transID : {duplicates:,}")
print(f"   Uniqueness Rate   : {((len(df_silver) - duplicates) / len(df_silver) * 1

# 3. Validity Check
print("\n3️⃣ VALIDITY CHECK")
print(f"   payAmount < 0     : {(df_silver['payAmount'] < 0).sum()}")
print(f"   direction values  : {sorted(df_silver['direction'].dropna().unique())}")
invalid_lat = ((df_silver['tapInStopsLat'].abs() > 90) | (df_silver['tapOutStopsLat
print(f"   Invalid Latitude  : {invalid_lat}")

# 4. Consistency Check
print("\n4️⃣ CONSISTENCY CHECK")
print(f"   payCardSex values : {df_silver['payCardSex'].unique()}")
print(f"   payCardBank unique: {df_silver['payCardBank'].nunique()}")
```

```
📊 DATA QUALITY CHECK — SEBELUM CLEANSING
============================================================

1️⃣ COMPLETENESS CHECK
   Total cells       : 4,169,000
   Missing cells     : 77,436
   Completeness Rate : 98.14%

2️⃣ UNIQUENESS CHECK
   Duplicate transID : 0
   Uniqueness Rate   : 100.00%

3️⃣ VALIDITY CHECK
   payAmount < 0     : 0
   direction values  : [0.0, 1.0]
   Invalid Latitude  : 0

4️⃣ CONSISTENCY CHECK
   payCardSex values : ['M' 'F']
   payCardBank unique: 6
```

In [9]:

```python
# === SILVER LAYER — Step 2: Data Cleansing ===
print("🧹 DATA CLEANSING PROCESS")
print("=" * 60)
```

```python
# 2a. Remove exact duplicates
before = len(df_silver)
df_silver = df_silver.drop_duplicates(subset=['transID'], keep='first')
print(f"\n✅ Removed {before - len(df_silver):,} duplicate rows")

# 2b. Fix data types — datetime columns
df_silver['tapInTime'] = pd.to_datetime(df_silver['tapInTime'], errors='coerce')
df_silver['tapOutTime'] = pd.to_datetime(df_silver['tapOutTime'], errors='coerce')
print("✅ Converted tapInTime & tapOutTime to datetime")

# 2c. Fix payCardBirthDate — convert to age
current_year = datetime.now().year
df_silver['payCardBirthDate'] = pd.to_numeric(df_silver['payCardBirthDate'], errors
df_silver['age'] = current_year - df_silver['payCardBirthDate']
# Cap unrealistic ages
df_silver.loc[df_silver['age'] < 0, 'age'] = np.nan
df_silver.loc[df_silver['age'] > 120, 'age'] = np.nan
print("✅ Created 'age' column from payCardBirthDate")

# 2d. Extract time features
df_silver['tapInHour'] = df_silver['tapInTime'].dt.hour
df_silver['tapInDate'] = df_silver['tapInTime'].dt.date
df_silver['tapInDayOfWeek'] = df_silver['tapInTime'].dt.day_name()
print("✅ Extracted time features: tapInHour, tapInDate, tapInDayOfWeek")

# 2e. Handle missing values
# corridorID & corridorName — fill with 'Unknown'
df_silver['corridorID'] = df_silver['corridorID'].fillna('Unknown')
df_silver['corridorName'] = df_silver['corridorName'].fillna('Unknown')
print("✅ Filled missing corridorID & corridorName with 'Unknown'")

# tapInStops & tapOutStops — fill with 'Unknown'
df_silver['tapInStops'] = df_silver['tapInStops'].fillna('Unknown')
df_silver['tapOutStops'] = df_silver['tapOutStops'].fillna('Unknown')
df_silver['tapOutStopsName'] = df_silver['tapOutStopsName'].fillna('Unknown')
print("✅ Filled missing stop columns with 'Unknown'")

# tapOutStopsLat/Lon — fill with 0
df_silver['tapOutStopsLat'] = df_silver['tapOutStopsLat'].fillna(0)
df_silver['tapOutStopsLon'] = df_silver['tapOutStopsLon'].fillna(0)
print("✅ Filled missing coordinates with 0")

# stopEndSeq — fill with 0
df_silver['stopEndSeq'] = df_silver['stopEndSeq'].fillna(0).astype(int)
print("✅ Filled missing stopEndSeq with 0")

# payAmount — fill with median
pay_median = df_silver['payAmount'].median()
df_silver['payAmount'] = df_silver['payAmount'].fillna(pay_median)
print(f"✅ Filled missing payAmount with median ({pay_median:,.0f})")

# 2f. Create age group
bins = [0, 17, 25, 35, 45, 55, 65, 120]
labels = ['<18', '18-25', '26-35', '36-45', '46-55', '56-65', '65+']
df_silver['ageGroup'] = pd.cut(df_silver['age'], bins=bins, labels=labels)
```

```
print("✅ Created 'ageGroup' column")

# Add cleaned timestamp
df_silver['_cleaned_timestamp'] = datetime.now().strftime('%Y-%m-%d %H:%M:%S')

print(f"\n📊 Silver Layer: {df_silver.shape[0]:,} rows × {df_silver.shape[1]} colu
```

```
🧹 DATA CLEANSING PROCESS
============================================================

✅ Removed 0 duplicate rows
✅ Converted tapInTime & tapOutTime to datetime
✅ Created 'age' column from payCardBirthDate
✅ Extracted time features: tapInHour, tapInDate, tapInDayOfWeek
✅ Filled missing corridorID & corridorName with 'Unknown'
✅ Filled missing stop columns with 'Unknown'
✅ Filled missing coordinates with 0
✅ Filled missing stopEndSeq with 0
✅ Filled missing payAmount with median (3,500)
✅ Created 'ageGroup' column

📊 Silver Layer: 189,500 rows × 33 columns
```

In [10]:

```
# === DATA QUALITY CHECK — SETELAH CLEANSING ===
print("📊 DATA QUALITY CHECK — SETELAH CLEANSING")
print("=" * 60)

check_cols = [c for c in df_silver.columns if not c.startswith('_')]

# 1. Completeness
total_cells = df_silver[check_cols].shape[0] * df_silver[check_cols].shape[1]
missing_cells = df_silver[check_cols].isnull().sum().sum()
completeness_after = ((total_cells - missing_cells) / total_cells * 100)
print(f"\n1️⃣ Completeness Rate : {completeness_after:.2f}% (sebelumnya: {completene

# 2. Uniqueness
dup_after = df_silver.duplicated(subset=['transID']).sum()
print(f"2️⃣ Duplicate transID : {dup_after}")

# 3. Missing values remaining
remaining_missing = df_silver[check_cols].isnull().sum()
remaining_missing = remaining_missing[remaining_missing > 0]
if len(remaining_missing) > 0:
    print(f"3️⃣ Remaining missing values:")
    for col, val in remaining_missing.items():
        print(f"   {col}: {val:,} ({val/len(df_silver)*100:.2f}%)")
else:
    print("3️⃣ No significant missing values remaining ✅")

# Quality Score
quality_score = completeness_after
df_silver['_quality_score'] = round(quality_score, 2)
print(f"\n⭐ Overall Data Quality Score: {quality_score:.2f}%")
```

```
📊 DATA QUALITY CHECK — SETELAH CLEANSING
============================================================

1️⃣ Completeness Rate : 99.87% (sebelumnya: 98.14%)
2️⃣ Duplicate transID : 0
3️⃣ Remaining missing values:
   tapOutTime: 6,720 (3.55%)

⭐ Overall Data Quality Score: 99.87%
```

🔒 Privacy Policy — Data Masking

Apa itu Privacy Policy dalam Data Engineering?

Privacy Policy adalah kebijakan perlindungan data yang memastikan bahwa **informasi pribadi (PII - Personally Identifiable Information)** dilindungi dari akses yang tidak sah.

Kolom PII yang akan di-mask:

Kolom	Tipe PII	Metode Masking
`payCardID`	Nomor kartu	Tampilkan 4 digit terakhir saja (`****5671`)
`payCardName`	Nama penumpang	Partial masking (`A*** S***`)
`payCardBirthDate`	Tahun lahir	Diganti dengan age group (sudah dilakukan)

Mengapa Data Masking penting?

1. **Kepatuhan regulasi** (GDPR, UU PDP Indonesia)
2. **Mencegah kebocoran data** sensitif
3. **Memungkinkan analisis** tanpa mengekspos identitas individu

---

In [11]:
```python
# === PRIVACY POLICY — DATA MASKING ===
print("🔒 DATA MASKING PROCESS")
print("=" * 60)

def mask_card_id(card_id):
    """Mask kartu: tampilkan 4 digit terakhir"""
    s = str(card_id)
    if len(s) <= 4:
        return '****'
    return '*' * (len(s) - 4) + s[-4:]

def mask_name(name):
    """Mask nama: tampilkan huruf pertama setiap kata"""
    if pd.isna(name) or str(name).strip() == '':
        return '***'
    words = str(name).split()
    masked = []
```

```python
    for w in words:
        if len(w) <= 1:
            masked.append(w[0] + '***')
        else:
            masked.append(w[0] + '*' * (len(w) - 1))
    return ' '.join(masked)

# Show before masking
print("\n📋 SEBELUM Masking:")
print(df_silver[['payCardID', 'payCardName', 'payCardBirthDate']].head(5).to_string

# Apply masking
df_silver['payCardID_masked'] = df_silver['payCardID'].apply(mask_card_id)
df_silver['payCardName_masked'] = df_silver['payCardName'].apply(mask_name)

# Replace original columns with masked versions
df_silver['payCardID_original'] = df_silver['payCardID']  # Keep backup
df_silver['payCardName_original'] = df_silver['payCardName']  # Keep backup
df_silver['payCardID'] = df_silver['payCardID_masked']
df_silver['payCardName'] = df_silver['payCardName_masked']

# Drop helper columns from export
df_silver.drop(columns=['payCardID_masked', 'payCardName_masked'], inplace=True)

print("\n📋 SETELAH Masking:")
print(df_silver[['payCardID', 'payCardName', 'ageGroup']].head(5).to_string())
print("\n✅ Data masking berhasil diterapkan!")
print("   - payCardID    → hanya 4 digit terakhir terlihat")
print("   - payCardName  → hanya huruf pertama setiap kata terlihat")
print("   - payCardBirthDate → diganti dengan ageGroup")
```

```
🔒 DATA MASKING PROCESS
============================================================

📋 SEBELUM Masking:
          payCardID                     payCardName  payCardBirthDate
0  3561407960318444     Dr. Janet Nashiruddin, M.Ak              2010
1   347728053419394               Balamantri Rahayu              2002
2   377105453850671                    Dian Mustofa              1993
3  4486493302356581        Cut Janet Suryatmi, M.Ak              1980
4    30139379978125  dr. Mulyanto Pudjiastuti, M.Ak              1997

📋 SETELAH Masking:
          payCardID                     payCardName ageGroup
0  ************8444     D** J**** N*********** M***      <18
1   ***********9394               B********* R*****    18-25
2   ***********0671                    D*** M******    26-35
3  ************6581        C** J**** S******** M***    46-55
4    **********8125  d** M******* P*********** M***    26-35

✅ Data masking berhasil diterapkan!
   - payCardID    → hanya 4 digit terakhir terlihat
   - payCardName  → hanya huruf pertama setiap kata terlihat
   - payCardBirthDate → diganti dengan ageGroup
```

🥇 Gold Layer — Aggregated & Analytics-Ready Data

Apa itu Gold Layer?

Gold Layer adalah **lapisan terakhir** dalam Medallion Architecture yang berisi data yang sudah **di-agregasi** dan **siap digunakan** untuk analisis, reporting, atau machine learning.

Karakteristik Gold Layer:

- Data sudah bersih dan tervalidasi
- Sudah di-agregasi sesuai kebutuhan bisnis
- Optimized untuk query dan reporting
- Tidak mengandung PII (data sudah di-mask)

Pada project ini, kita membuat beberapa **Gold Tables**:

1. **Transaksi per Corridor** — revenue dan jumlah transaksi per corridor
2. **Traffic per Jam** — distribusi transaksi per jam
3. **Demografi Penumpang** — statistik berdasarkan gender dan age group
4. **Daily Trend** — trend transaksi harian

In [12]:

```
# === GOLD LAYER – Aggregated Tables ===
print("🥇 GOLD LAYER – Creating Aggregated Tables")
print("=" * 60)

# Gold Table 1: Corridor Summary
gold_corridor = df_silver.groupby(['corridorID', 'corridorName']).agg(
    total_transactions=('transID', 'count'),
    total_revenue=('payAmount', 'sum'),
    avg_revenue=('payAmount', 'mean'),
    unique_passengers=('payCardID', 'nunique')
).reset_index().sort_values('total_transactions', ascending=False)
gold_corridor['_aggregation_timestamp'] = datetime.now().strftime('%Y-%m-%d %H:%M:%
gold_corridor['_layer'] = 'gold'
print(f"✅ Gold Table 1 – Corridor Summary: {gold_corridor.shape[0]} rows")

# Gold Table 2: Hourly Traffic
gold_hourly = df_silver.groupby('tapInHour').agg(
    total_transactions=('transID', 'count'),
    total_revenue=('payAmount', 'sum'),
    avg_revenue=('payAmount', 'mean')
).reset_index()
gold_hourly['_aggregation_timestamp'] = datetime.now().strftime('%Y-%m-%d %H:%M:%S'
gold_hourly['_layer'] = 'gold'
print(f"✅ Gold Table 2 – Hourly Traffic: {gold_hourly.shape[0]} rows")

# Gold Table 3: Demographics Summary
gold_demographics = df_silver.groupby(['payCardSex', 'ageGroup']).agg(
```

```
    total_transactions=('transID', 'count'),
    total_revenue=('payAmount', 'sum'),
    avg_revenue=('payAmount', 'mean')
).reset_index()
gold_demographics['_aggregation_timestamp'] = datetime.now().strftime('%Y-%m-%d %H:
gold_demographics['_layer'] = 'gold'
print(f"✅ Gold Table 3 — Demographics: {gold_demographics.shape[0]} rows")

# Gold Table 4: Daily Trend
gold_daily = df_silver.groupby('tapInDate').agg(
    total_transactions=('transID', 'count'),
    total_revenue=('payAmount', 'sum'),
    unique_passengers=('payCardID', 'nunique')
).reset_index()
gold_daily['tapInDate'] = pd.to_datetime(gold_daily['tapInDate'])
gold_daily['_aggregation_timestamp'] = datetime.now().strftime('%Y-%m-%d %H:%M:%S')
gold_daily['_layer'] = 'gold'
print(f"✅ Gold Table 4 — Daily Trend: {gold_daily.shape[0]} rows")

print(f"\n📊 Total Gold Tables: 4")
```

```
🥇 GOLD LAYER — Creating Aggregated Tables
============================================================
✅ Gold Table 1 — Corridor Summary: 655 rows
✅ Gold Table 2 — Hourly Traffic: 17 rows
✅ Gold Table 3 — Demographics: 14 rows
✅ Gold Table 4 — Daily Trend: 30 rows

📊 Total Gold Tables: 4
```

In [13]:

```
# Preview Gold Tables
print("📋 PREVIEW GOLD TABLES")
print("=" * 60)

print("\n🏆 Top 10 Corridors by Transactions:")
print(gold_corridor[['corridorName', 'total_transactions', 'total_revenue', 'unique

print("\n⏰ Hourly Traffic:")
print(gold_hourly[['tapInHour', 'total_transactions', 'total_revenue']].to_string(i
```

```
📋 PREVIEW GOLD TABLES
============================================================

🏆 Top 10 Corridors by Transactions:
                        corridorName  total_transactions  total_revenue  unique_pass
engers
                             Unknown                5707    15518000.00
3426
     Rusun Penjaringan - Penjaringan                1207     4224500.00
96
              Pulo Gebang - Matraman                1183     4140500.00
108
                   Ciledug - Tendean                1150     4025000.00
99
            Kalimalang - Duren Sawit                1121           0.00
101
                 Bintaro - Fatmawati                1101    22020000.00
95
               Matraman Baru - Ancol                1099     3846500.00
113
      Kampung Rambutan - Bidara Cina                1095     3832500.00
104
Rusun Pinus Elok - Rusun Pulo Gebang                1078           0.00
101
                Harapan Indah - ASMI                1074     3759000.00
86

⏰ Hourly Traffic:
 tapInHour  total_transactions  total_revenue
         5               16807    45130500.00
         6               29133    77995000.00
         7               14194    37784500.00
         8               14314    39082500.00
         9               14043    37647500.00
        10                1691     4743500.00
        11                1680     4399500.00
        12                1732     4687500.00
        13                1735     4885000.00
        14                 913     2490000.00
        15                2073     5926000.00
        16               17137    45512500.00
        17               27096    72707500.00
        18               11930    31936500.00
        19               12100    32868500.00
        20               11957    32300000.00
        21               10965    29553000.00
```

📚 Metadata Strategy — Data Dictionary

Apa itu Metadata Strategy?

Metadata Strategy adalah pendekatan terstruktur untuk mendokumentasikan, mengelola, dan memanfaatkan metadata (data tentang data) dalam suatu organisasi.

Apa itu Data Dictionary?

Data Dictionary adalah dokumen yang menjelaskan struktur, definisi, dan atribut dari setiap elemen data dalam dataset. Ini berfungsi sebagai **referensi utama** bagi semua pengguna data.

Manfaat Data Dictionary:

- Memastikan **pemahaman yang seragam** tentang setiap kolom
- Mempermudah **onboarding** anggota tim baru
- Mendukung **audit** dan **compliance**
- Menjadi bagian dari **data catalog** organisasi

In [14]:

```
# === DATA DICTIONARY ===
print("📚 DATA DICTIONARY — Silver Layer")
print("=" * 80)

data_dict = [
    ['transID', 'object', 'ID unik transaksi', 'No', 'No'],
    ['payCardID', 'object', 'ID kartu pembayaran (MASKED)', 'No', 'Yes ⚠️'],
    ['payCardBank', 'object', 'Bank penerbit kartu', 'No', 'No'],
    ['payCardName', 'object', 'Nama pemegang kartu (MASKED)', 'No', 'Yes ⚠️'],
    ['payCardSex', 'object', 'Jenis kelamin pemegang kartu (F/M)', 'No', 'No'],
    ['payCardBirthDate', 'int64', 'Tahun lahir pemegang kartu', 'No', 'Yes ⚠️'],
    ['corridorID', 'object', 'ID koridor TransJakarta', 'No', 'No'],
    ['corridorName', 'object', 'Nama koridor TransJakarta', 'No', 'No'],
    ['direction', 'float64', 'Arah perjalanan (0=Pergi, 1=Pulang)', 'No', 'No'],
    ['tapInStops', 'object', 'ID halte tap in', 'No', 'No'],
    ['tapInStopsName', 'object', 'Nama halte tap in', 'No', 'No'],
    ['tapInStopsLat', 'float64', 'Latitude halte tap in', 'No', 'No'],
    ['tapInStopsLon', 'float64', 'Longitude halte tap in', 'No', 'No'],
    ['stopStartSeq', 'int64', 'Urutan halte tap in', 'No', 'No'],
    ['tapInTime', 'datetime64', 'Waktu tap in', 'No', 'No'],
    ['tapOutStops', 'object', 'ID halte tap out', 'Yes', 'No'],
    ['tapOutStopsName', 'object', 'Nama halte tap out', 'Yes', 'No'],
    ['tapOutStopsLat', 'float64', 'Latitude halte tap out', 'Yes', 'No'],
    ['tapOutStopsLon', 'float64', 'Longitude halte tap out', 'Yes', 'No'],
    ['stopEndSeq', 'int64', 'Urutan halte tap out', 'Yes', 'No'],
    ['tapOutTime', 'datetime64', 'Waktu tap out', 'Yes', 'No'],
    ['payAmount', 'float64', 'Jumlah pembayaran (Rp)', 'No', 'No'],
    ['age', 'float64', 'Umur penumpang (kalkulasi)', 'Yes', 'No'],
    ['ageGroup', 'category', 'Kelompok umur penumpang', 'Yes', 'No'],
    ['tapInHour', 'float64', 'Jam tap in (0-23)', 'Yes', 'No'],
    ['tapInDate', 'object', 'Tanggal tap in', 'Yes', 'No'],
    ['tapInDayOfWeek', 'object', 'Hari dalam seminggu', 'Yes', 'No'],
]

dd_df = pd.DataFrame(data_dict, columns=['Column Name', 'Data Type', 'Description',
print(dd_df.to_string(index=False))
print(f"\n⚠️  Kolom dengan PII Flag telah di-mask pada Silver Layer")
```

```
📚 DATA DICTIONARY — Silver Layer
================================================================================
     Column Name  Data Type                        Description Nullable PII Flag
         transID     object                   ID unik transaksi       No       No
       payCardID     object        ID kartu pembayaran (MASKED)       No   Yes ⚠️
     payCardBank     object                 Bank penerbit kartu       No       No
     payCardName     object        Nama pemegang kartu (MASKED)       No   Yes ⚠️
      payCardSex     object  Jenis kelamin pemegang kartu (F/M)       No       No
payCardBirthDate      int64          Tahun lahir pemegang kartu       No   Yes ⚠️
      corridorID     object             ID koridor TransJakarta       No       No
    corridorName     object           Nama koridor TransJakarta       No       No
       direction    float64 Arah perjalanan (0=Pergi, 1=Pulang)       No       No
      tapInStops     object                     ID halte tap in       No       No
  tapInStopsName     object                   Nama halte tap in       No       No
   tapInStopsLat    float64               Latitude halte tap in       No       No
   tapInStopsLon    float64              Longitude halte tap in       No       No
    stopStartSeq      int64                 Urutan halte tap in       No       No
       tapInTime datetime64                        Waktu tap in       No       No
     tapOutStops     object                    ID halte tap out      Yes       No
 tapOutStopsName     object                  Nama halte tap out      Yes       No
  tapOutStopsLat    float64              Latitude halte tap out      Yes       No
  tapOutStopsLon    float64             Longitude halte tap out      Yes       No
      stopEndSeq      int64                Urutan halte tap out      Yes       No
      tapOutTime datetime64                       Waktu tap out      Yes       No
       payAmount    float64              Jumlah pembayaran (Rp)       No       No
             age    float64          Umur penumpang (kalkulasi)      Yes       No
        ageGroup   category             Kelompok umur penumpang      Yes       No
       tapInHour    float64                   Jam tap in (0-23)      Yes       No
       tapInDate     object                      Tanggal tap in      Yes       No
  tapInDayOfWeek     object                 Hari dalam seminggu      Yes       No

⚠️  Kolom dengan PII Flag telah di-mask pada Silver Layer
```

📊 Data Analysis & Visualization

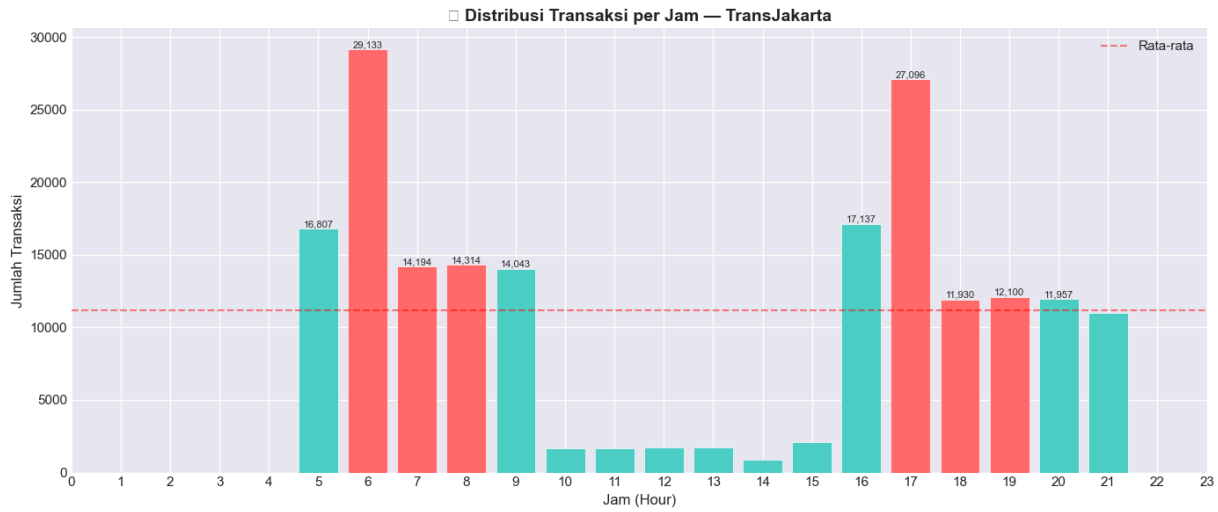
Berikut adalah analisis data lengkap beserta visualisasi dan insight dari dataset Transjakarta.

1️⃣ Distribusi Transaksi per Jam

In [15]:

```
fig, ax = plt.subplots(figsize=(14, 6))
colors = ['#FF6B6B' if h in [6,7,8,17,18,19] else '#4ECDC4' for h in gold_hourly['t
bars = ax.bar(gold_hourly['tapInHour'], gold_hourly['total_transactions'], color=co
ax.set_xlabel('Jam (Hour)', fontsize=12)
ax.set_ylabel('Jumlah Transaksi', fontsize=12)
ax.set_title('📊 Distribusi Transaksi per Jam — TransJakarta', fontsize=14, fontwe
ax.set_xticks(range(0, 24))
ax.axhline(y=gold_hourly['total_transactions'].mean(), color='red', linestyle='--',
ax.legend()
for bar in bars:
    height = bar.get_height()
    if height > gold_hourly['total_transactions'].mean():
        ax.text(bar.get_x() + bar.get_width()/2., height, f'{int(height):,}', ha='c
```

```
plt.tight_layout()
plt.show()

print("\n💡 INSIGHT:")
peak_hour = gold_hourly.loc[gold_hourly['total_transactions'].idxmax()]
print(f"   • Jam tersibuk: {int(peak_hour['tapInHour']):02d}:00 dengan {int(peak_ho
print(f"   • Jam sibuk pagi (06-09) dan sore (17-19) menunjukkan pola commuter/peke
print(f"   • Warna merah = jam sibuk (rush hour), biru = jam normal")
```

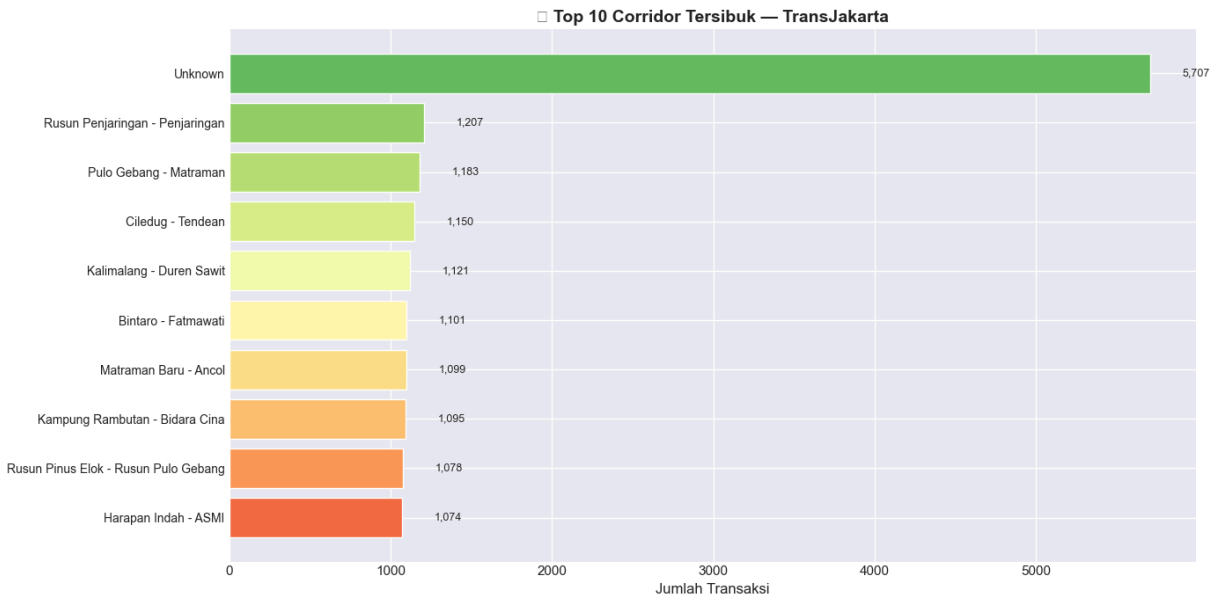



```
💡 INSIGHT:
   • Jam tersibuk: 06:00 dengan 29,133 transaksi
   • Jam sibuk pagi (06-09) dan sore (17-19) menunjukkan pola commuter/pekerja
   • Warna merah = jam sibuk (rush hour), biru = jam normal
```

2️⃣ Top 10 Corridor Tersibuk

In [16]:

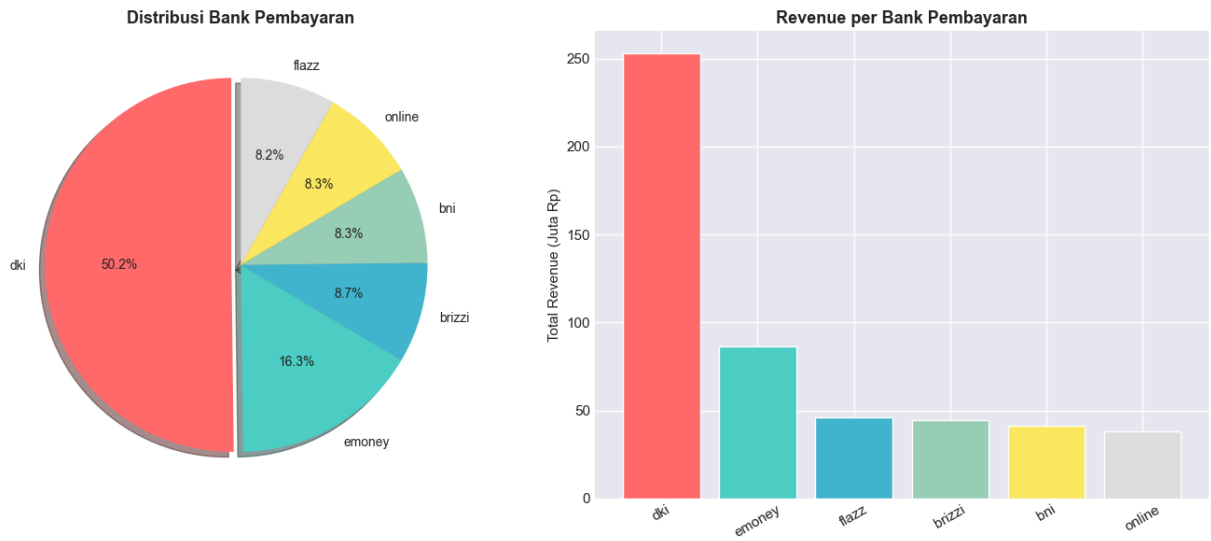
```
fig, ax = plt.subplots(figsize=(14, 7))
top10 = gold_corridor.head(10)
colors = plt.cm.RdYlGn_r(np.linspace(0.2, 0.8, 10))
bars = ax.barh(range(len(top10)), top10['total_transactions'], color=colors, edgeco
ax.set_yticks(range(len(top10)))
ax.set_yticklabels(top10['corridorName'].str[:40], fontsize=10)
ax.set_xlabel('Jumlah Transaksi', fontsize=12)
ax.set_title('🏆 Top 10 Corridor Tersibuk — TransJakarta', fontsize=14, fontweight=
ax.invert_yaxis()
for i, bar in enumerate(bars):
    width = bar.get_width()
    ax.text(width + 200, bar.get_y() + bar.get_height()/2, f'{int(width):,}', va='c
plt.tight_layout()
plt.show()

print("\n💡 INSIGHT:")
print(f"   • Corridor tersibuk: {top10.iloc[0]['corridorName']}")
print(f"     dengan {int(top10.iloc[0]['total_transactions']):,} transaksi")
print(f"   • Top 3 corridor menguasai {top10.head(3)['total_transactions'].sum()/go
```




```
💡 INSIGHT:
  • Corridor tersibuk: Unknown
    dengan 5,707 transaksi
  • Top 3 corridor menguasai 4.3% total transaksi
```

3️⃣ Distribusi Metode Pembayaran (Bank)

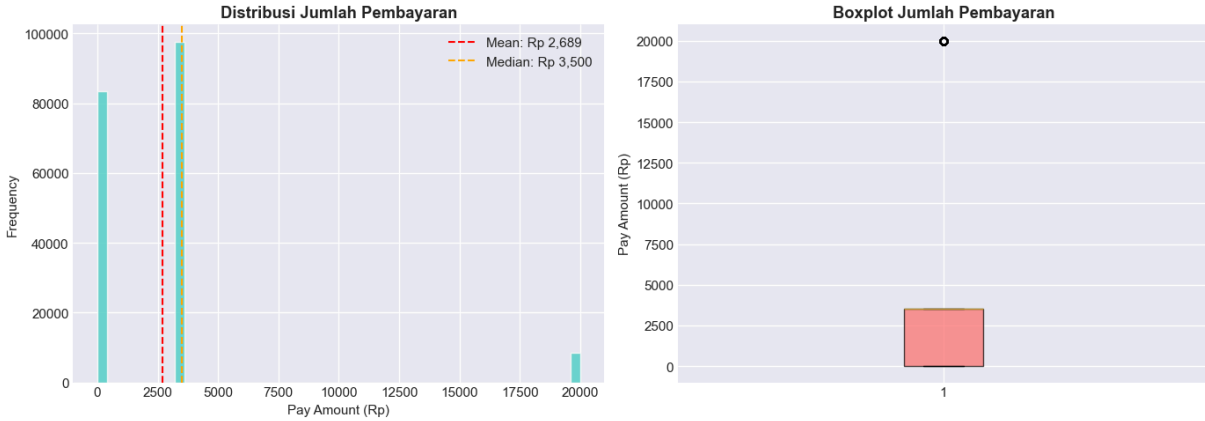
In [17]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# Pie chart
bank_dist = df_silver['payCardBank'].value_counts().head(6)
colors = ['#FF6B6B', '#4ECDC4', '#45B7D1', '#96CEB4', '#FCEA61', '#DDD']
explode = [0.05] + [0] * (len(bank_dist) - 1)
axes[0].pie(bank_dist, labels=bank_dist.index, autopct='%1.1f%%', colors=colors,
            explode=explode, shadow=True, startangle=90, textprops={'fontsize': 10}
axes[0].set_title('Distribusi Bank Pembayaran', fontsize=13, fontweight='bold')

# Bar chart
bank_revenue = df_silver.groupby('payCardBank')['payAmount'].sum().sort_values(asce
axes[1].bar(bank_revenue.index, bank_revenue.values / 1e6, color=colors, edgecolor=
axes[1].set_ylabel('Total Revenue (Juta Rp)', fontsize=11)
axes[1].set_title('Revenue per Bank Pembayaran', fontsize=13, fontweight='bold')
axes[1].tick_params(axis='x', rotation=30)

plt.tight_layout()
plt.show()

print("\n💡 INSIGHT:")
print(f"   • Bank paling populer: {bank_dist.index[0]} ({bank_dist.values[0]:,} tra
print(f"   • {bank_dist.index[0]} juga menghasilkan revenue tertinggi")
```



💡 INSIGHT:
- Bank paling populer: dki (95,126 transaksi, 50.2%)
- dki juga menghasilkan revenue tertinggi

4️⃣ Distribusi Jumlah Pembayaran

In [18]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Histogram
axes[0].hist(df_silver['payAmount'], bins=50, color='#4ECDC4', edgecolor='white', a
axes[0].axvline(df_silver['payAmount'].mean(), color='red', linestyle='--', label=f
axes[0].axvline(df_silver['payAmount'].median(), color='orange', linestyle='--', la
axes[0].set_xlabel('Pay Amount (Rp)')
axes[0].set_ylabel('Frequency')
axes[0].set_title('Distribusi Jumlah Pembayaran', fontweight='bold')
axes[0].legend()

# Boxplot
axes[1].boxplot(df_silver['payAmount'].dropna(), vert=True, patch_artist=True,
                boxprops=dict(facecolor='#FF6B6B', alpha=0.7))
axes[1].set_ylabel('Pay Amount (Rp)')
axes[1].set_title('Boxplot Jumlah Pembayaran', fontweight='bold')

plt.tight_layout()
plt.show()

print("\n💡 INSIGHT:")
print(f"   • Rata-rata pembayaran: Rp {df_silver['payAmount'].mean():,.0f}")
print(f"   • Median pembayaran: Rp {df_silver['payAmount'].median():,.0f}")
free_pct = (df_silver['payAmount'] == 0).sum() / len(df_silver) * 100
print(f"   • Transaksi gratis (Rp 0): {free_pct:.1f}% — kemungkinan penumpang denga
```



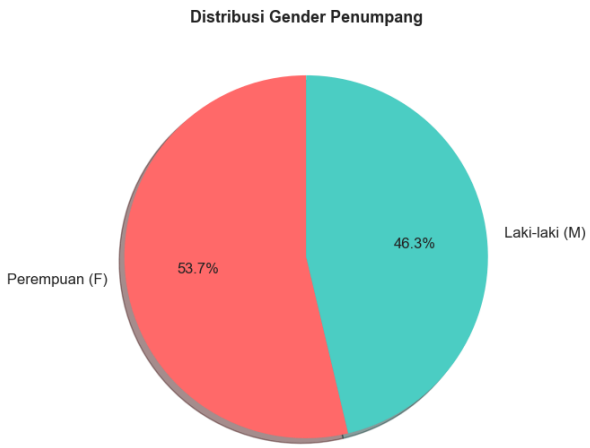
💡 INSIGHT:

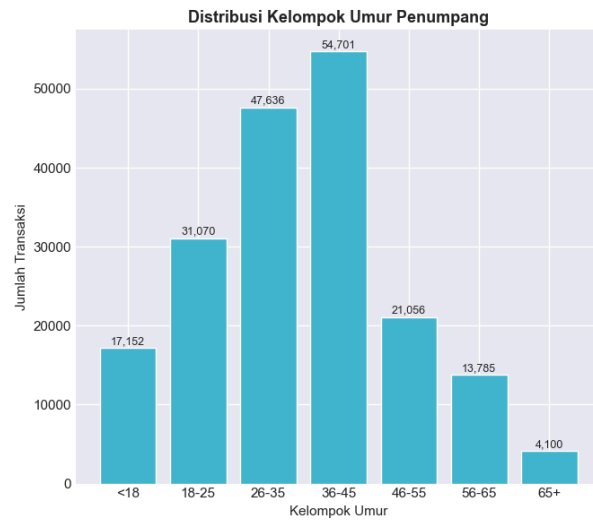
- Rata-rata pembayaran: Rp 2,689
- Median pembayaran: Rp 3,500
- Transaksi gratis (Rp 0): 44.0% — kemungkinan penumpang dengan kartu subsidi

5️⃣ Demografi Penumpang

In [19]:

```
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# Gender distribution
gender_counts = df_silver['payCardSex'].value_counts()
colors_g = ['#FF6B6B', '#4ECDC4']
axes[0].pie(gender_counts, labels=['Perempuan (F)', 'Laki-laki (M)'] if gender_coun
            autopct='%1.1f%%', colors=colors_g, shadow=True, startangle=90, textpro
axes[0].set_title('Distribusi Gender Penumpang', fontsize=13, fontweight='bold')

# Age group distribution
age_counts = df_silver['ageGroup'].value_counts().sort_index()
axes[1].bar(age_counts.index.astype(str), age_counts.values, color='#45B7D1', edgec
axes[1].set_xlabel('Kelompok Umur')
axes[1].set_ylabel('Jumlah Transaksi')
axes[1].set_title('Distribusi Kelompok Umur Penumpang', fontsize=13, fontweight='bo
for i, v in enumerate(age_counts.values):
    axes[1].text(i, v + 500, f'{v:,}', ha='center', fontsize=9)

plt.tight_layout()
plt.show()

print("\n💡 INSIGHT:")
print(f"   • Gender dominan: {gender_counts.index[0]} ({gender_counts.values[0]:,}
dominant_age = age_counts.idxmax()
print(f"   • Kelompok umur terbanyak: {dominant_age} ({age_counts.max():,} transaks
print(f"   • Mayoritas penumpang adalah usia produktif — sesuai pola commuter")
```





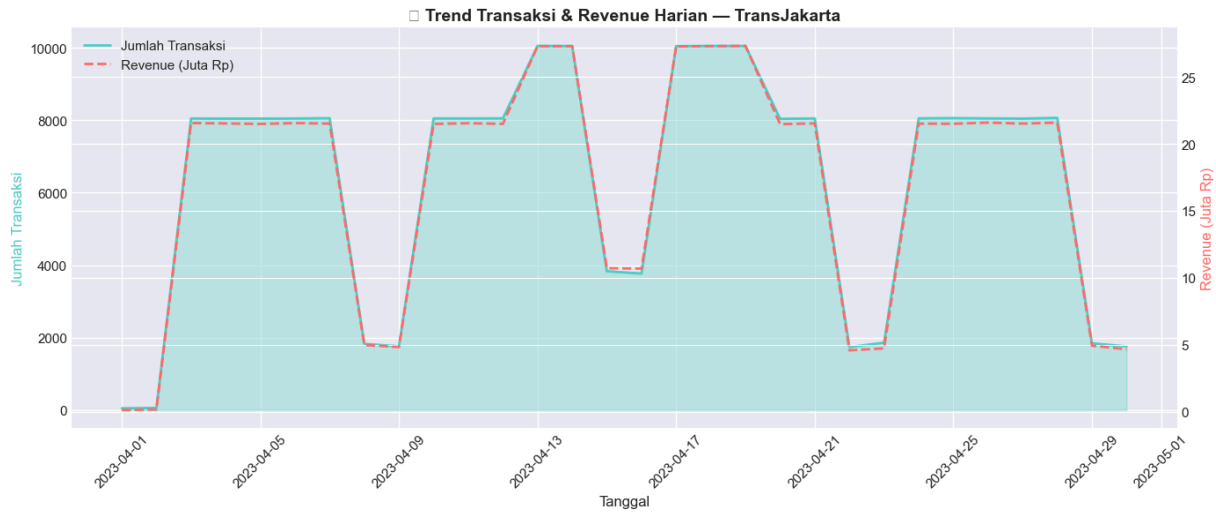
💡 INSIGHT:

- Gender dominan: F (101,790 transaksi)
- Kelompok umur terbanyak: 36-45 (54,701 transaksi)
- Mayoritas penumpang adalah usia produktif — sesuai pola commuter

6️⃣ Trend Transaksi Harian

In [20]:

```
fig, ax1 = plt.subplots(figsize=(14, 6))

ax1.fill_between(gold_daily['tapInDate'], gold_daily['total_transactions'], alpha=0
ax1.plot(gold_daily['tapInDate'], gold_daily['total_transactions'], color='#4ECDC4'
ax1.set_xlabel('Tanggal', fontsize=12)
ax1.set_ylabel('Jumlah Transaksi', fontsize=12, color='#4ECDC4')
ax1.tick_params(axis='x', rotation=45)

ax2 = ax1.twinx()
ax2.plot(gold_daily['tapInDate'], gold_daily['total_revenue']/1e6, color='#FF6B6B',
ax2.set_ylabel('Revenue (Juta Rp)', fontsize=12, color='#FF6B6B')

lines1, labels1 = ax1.get_legend_handles_labels()
lines2, labels2 = ax2.get_legend_handles_labels()
ax1.legend(lines1 + lines2, labels1 + labels2, loc='upper left')

ax1.set_title('📈 Trend Transaksi & Revenue Harian — TransJakarta', fontsize=14, f
plt.tight_layout()
plt.show()

print("\n💡 INSIGHT:")
busiest_day = gold_daily.loc[gold_daily['total_transactions'].idxmax()]
print(f"   • Hari tersibuk: {busiest_day['tapInDate'].strftime('%Y-%m-%d')} ({int(b
print(f"   • Revenue tertinggi: Rp {gold_daily['total_revenue'].max():,.0f}")
print(f"   • Terdapat pola penurunan di akhir pekan (weekend)")
```

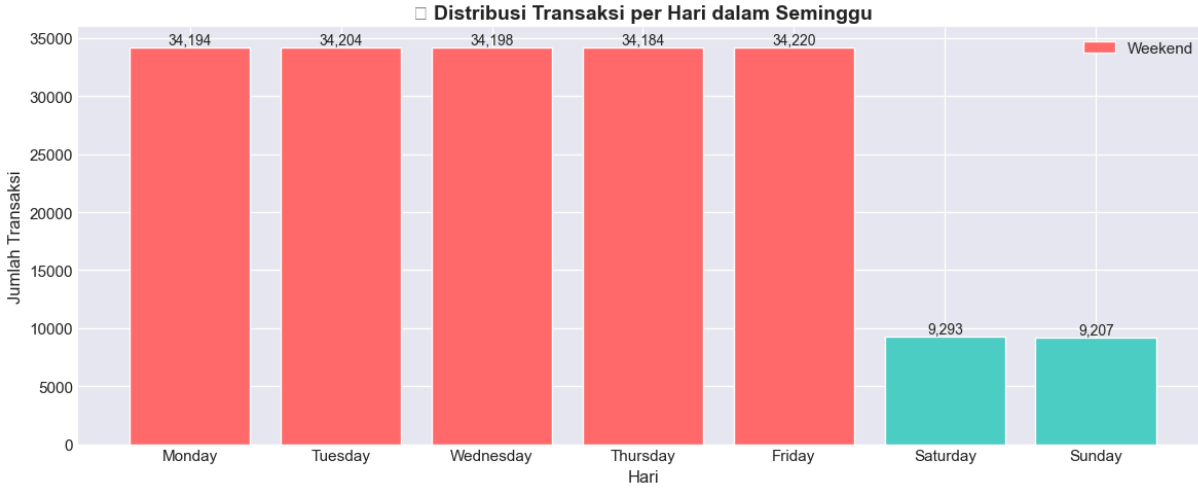


💡 INSIGHT:

- Hari tersibuk: 2023-04-13 (10,052 transaksi)
- Revenue tertinggi: Rp 27,316,000
- Terdapat pola penurunan di akhir pekan (weekend)

7️⃣ Distribusi Transaksi per Hari

In [21]:
```
day_order = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'S
day_counts = df_silver['tapInDayOfWeek'].value_counts().reindex(day_order)

fig, ax = plt.subplots(figsize=(12, 5))
colors = ['#4ECDC4' if d in ['Saturday', 'Sunday'] else '#FF6B6B' for d in day_orde
bars = ax.bar(day_counts.index, day_counts.values, color=colors, edgecolor='white')
ax.set_xlabel('Hari', fontsize=12)
ax.set_ylabel('Jumlah Transaksi', fontsize=12)
ax.set_title('📊 Distribusi Transaksi per Hari dalam Seminggu', fontsize=14, fontwe
for bar in bars:
    height = bar.get_height()
    ax.text(bar.get_x() + bar.get_width()/2., height, f'{int(height):,}', ha='cente
ax.legend(['Weekend', 'Weekday'], loc='upper right')
plt.tight_layout()
plt.show()

print("\n💡 INSIGHT:")
print(f"   • Transaksi weekday jauh lebih tinggi dari weekend — pola commuter")
wkday = day_counts[['Monday','Tuesday','Wednesday','Thursday','Friday']].mean()
wkend = day_counts[['Saturday','Sunday']].mean()
print(f"   • Rata-rata weekday: {wkday:,.0f} vs weekend: {wkend:,.0f}")
print(f"   • Selisih: {((wkday-wkend)/wkend*100):.0f}% lebih tinggi di weekday")
```



💡 INSIGHT:
- Transaksi weekday jauh lebih tinggi dari weekend — pola commuter
- Rata-rata weekday: 34,200 vs weekend: 9,250
- Selisih: 270% lebih tinggi di weekday

8️⃣ Peta Halte Populer

In [22]:
```
# Filter valid coordinates only
valid_coords = df_silver[(df_silver['tapInStopsLat'] != 0) & (df_silver['tapInStops
stop_traffic = valid_coords.groupby(['tapInStopsName', 'tapInStopsLat', 'tapInStops
stop_traffic = stop_traffic.sort_values('count', ascending=False)

fig, ax = plt.subplots(figsize=(14, 10))
scatter = ax.scatter(
    stop_traffic['tapInStopsLon'], stop_traffic['tapInStopsLat'],
    s=stop_traffic['count'] / stop_traffic['count'].max() * 300,
    c=stop_traffic['count'], cmap='YlOrRd', alpha=0.6, edgecolors='gray', linewidth
)
plt.colorbar(scatter, label='Jumlah Transaksi', shrink=0.8)

# Label top 10 stops
for _, row in stop_traffic.head(10).iterrows():
    ax.annotate(row['tapInStopsName'][:20], (row['tapInStopsLon'], row['tapInStopsL
                fontsize=7, ha='center', va='bottom', fontweight='bold',
                bbox=dict(boxstyle='round,pad=0.2', facecolor='white', alpha=0.7))

ax.set_xlabel('Longitude', fontsize=12)
ax.set_ylabel('Latitude', fontsize=12)
ax.set_title('🗺️ Peta Halte Populer — TransJakarta', fontsize=14, fontweight='bold
plt.tight_layout()
plt.show()

print("\n💡 INSIGHT:")
print(f"   • Total halte unik: {stop_traffic.shape[0]:,}")
print(f"   • Halte tersibuk:")
for i, row in stop_traffic.head(5).iterrows():
    print(f"     {row['tapInStopsName']}: {row['count']:,} transaksi")
```

💡 INSIGHT:

- Total halte unik: 5,278
- Halte tersibuk:
  Penjaringan: 1,396 transaksi
  BKN: 794 transaksi
  BNN LRT: 758 transaksi
  Rusun Penjaringan: 654 transaksi
  Garuda Taman Mini: 642 transaksi

💾 Export Data

Data yang sudah bersih akan di-export ke dua format:

1. **Excel (.xlsx)** — dengan multiple sheets untuk Silver Layer dan Gold Tables
2. **MySQL** — menggunakan SQLAlchemy + PyMySQL

> **Note:** Kolom PII asli (`payCardID_original` , `payCardName_original`) **TIDAK** di-export untuk menjaga privacy.

In [23]:
```
# === EXPORT TO EXCEL ===
print("💾 EXPORT TO EXCEL")
print("=" * 60)

# Prepare Silver data for export (remove internal & original PII columns)
export_exclude = ['_ingestion_timestamp', '_source_file', '_source_origin', '_recor
```

```
                    '_layer', '_cleaned_timestamp', '_quality_score',
                    'payCardID_original', 'payCardName_original']
silver_export = df_silver.drop(columns=[c for c in export_exclude if c in df_silver

# Convert date columns for Excel compatibility
if 'tapInDate' in silver_export.columns:
    silver_export['tapInDate'] = silver_export['tapInDate'].astype(str)

EXCEL_PATH = 'TransJakarta_Clean_Data.xlsx'

with pd.ExcelWriter(EXCEL_PATH, engine='openpyxl') as writer:
    silver_export.to_excel(writer, sheet_name='Silver_Layer', index=False)
    gold_corridor.to_excel(writer, sheet_name='Gold_Corridor', index=False)
    gold_hourly.to_excel(writer, sheet_name='Gold_Hourly', index=False)
    gold_demographics.to_excel(writer, sheet_name='Gold_Demographics', index=False)
    gold_daily.to_excel(writer, sheet_name='Gold_Daily', index=False)

print(f"✅ Exported to Excel: {EXCEL_PATH}")
print(f"   Sheets: Silver_Layer, Gold_Corridor, Gold_Hourly, Gold_Demographics, Gol

import os
file_size = os.path.getsize(EXCEL_PATH) / 1024 / 1024
print(f"   File size: {file_size:.2f} MB")
```

```
💾 EXPORT TO EXCEL
============================================================
✅ Exported to Excel: TransJakarta_Clean_Data.xlsx
   Sheets: Silver_Layer, Gold_Corridor, Gold_Hourly, Gold_Demographics, Gold_Daily
   File size: 36.45 MB
```

💾 Export ke File SQL (.sql)

File `.sql` ini bisa di-import secara manual ke MySQL menggunakan:

```
mysql -u root -p transjakarta_db < TransJakarta_Export.sql
```

Atau melalui phpMyAdmin / MySQL Workbench dengan fitur **Import**.

In [24]:

```
# === EXPORT TO SQL FILE ===
print("💾 EXPORT TO SQL FILE (.sql)")
print("=" * 60)

import os

def escape_sql(val):
    """Escape value untuk SQL statement"""
    if pd.isna(val):
        return 'NULL'
    if isinstance(val, (int, float, np.integer, np.floating)):
        if pd.isna(val):
            return 'NULL'
        return str(val)
    s = str(val).replace("\\", "\\\\").replace("'", "\\'")
    return f"'{s}'"
```

```python
def get_sql_type(dtype, col_name):
    """Map pandas dtype ke MySQL type"""
    dtype_str = str(dtype)
    if 'int' in dtype_str:
        return 'BIGINT'
    elif 'float' in dtype_str:
        return 'DOUBLE'
    elif 'datetime' in dtype_str:
        return 'DATETIME'
    elif 'date' in dtype_str:
        return 'DATE'
    elif 'bool' in dtype_str:
        return 'BOOLEAN'
    else:
        max_len = 255
        return f'VARCHAR({max_len})'

def df_to_sql_statements(df, table_name):
    """Generate CREATE TABLE + INSERT statements dari DataFrame"""
    statements = []

    # DROP TABLE IF EXISTS
    statements.append(f"DROP TABLE IF EXISTS `{table_name}`;")
    statements.append("")

    # CREATE TABLE
    col_defs = []
    for col in df.columns:
        sql_type = get_sql_type(df[col].dtype, col)
        col_defs.append(f"  `{col}` {sql_type}")
    create_stmt = f"CREATE TABLE `{table_name}` (\n"
    create_stmt += ",\n".join(col_defs)
    create_stmt += "\n) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;"
    statements.append(create_stmt)
    statements.append("")

    # INSERT statements (batch per 500 rows)
    batch_size = 500
    for start in range(0, len(df), batch_size):
        batch = df.iloc[start:start + batch_size]
        cols_str = ", ".join([f"`{c}`" for c in df.columns])
        insert = f"INSERT INTO `{table_name}` ({cols_str}) VALUES\n"
        rows = []
        for _, row in batch.iterrows():
            vals = ", ".join([escape_sql(row[c]) for c in df.columns])
            rows.append(f"({vals})")
        insert += ",\n".join(rows) + ";"
        statements.append(insert)
        statements.append("")

    return "\n".join(statements)

# Prepare export data (tanpa kolom internal & PII original)
export_exclude = ['_ingestion_timestamp', '_source_file', '_source_origin', '_recor
                  '_layer', '_cleaned_timestamp', '_quality_score',
```

```
                    'payCardID_original', 'payCardName_original']
silver_export = df_silver.drop(columns=[c for c in export_exclude if c in df_silver

# Convert datetime columns to string for SQL
for col in silver_export.columns:
    if silver_export[col].dtype == 'datetime64[ns]':
        silver_export[col] = silver_export[col].astype(str).replace('NaT', None)
if 'tapInDate' in silver_export.columns:
    silver_export['tapInDate'] = silver_export['tapInDate'].astype(str)

SQL_PATH = 'TransJakarta_Export.sql'

with open(SQL_PATH, 'w', encoding='utf-8') as f:
    # Header
    f.write("-- ==============================================\n")
    f.write("-- TransJakarta Data Engineer Export\n")
    f.write(f"-- Generated: {datetime.now().strftime('%Y-%m-%d %H:%M:%S')}\n")
    f.write("-- Database: transjakarta_db\n")
    f.write("-- ==============================================\n\n")
    f.write("CREATE DATABASE IF NOT EXISTS `transjakarta_db`;\n")
    f.write("USE `transjakarta_db`;\n\n")
    f.write("SET NAMES utf8mb4;\n")
    f.write("SET FOREIGN_KEY_CHECKS = 0;\n\n")

    # Silver Layer
    print("  Generating SQL for silver_transactions...")
    f.write("-- ----------------------------------------\n")
    f.write("-- Table: silver_transactions (Silver Layer)\n")
    f.write("-- ----------------------------------------\n\n")
    f.write(df_to_sql_statements(silver_export, 'silver_transactions'))
    f.write("\n\n")

    # Gold Tables
    gold_tables = {
        'gold_corridor_summary': gold_corridor,
        'gold_hourly_traffic': gold_hourly,
        'gold_demographics': gold_demographics,
        'gold_daily_trend': gold_daily,
    }

    for tbl_name, tbl_df in gold_tables.items():
        print(f"  Generating SQL for {tbl_name}...")
        tbl_export = tbl_df.copy()
        for col in tbl_export.columns:
            if tbl_export[col].dtype == 'datetime64[ns]':
                tbl_export[col] = tbl_export[col].astype(str).replace('NaT', None)
        f.write(f"-- ----------------------------------------\n")
        f.write(f"-- Table: {tbl_name} (Gold Layer)\n")
        f.write(f"-- ----------------------------------------\n\n")
        f.write(df_to_sql_statements(tbl_export, tbl_name))
        f.write("\n\n")

    f.write("SET FOREIGN_KEY_CHECKS = 1;\n")

file_size_mb = os.path.getsize(SQL_PATH) / 1024 / 1024
print(f"\n✅ Exported to SQL file: {SQL_PATH}")
```

```
print(f"   File size: {file_size_mb:.2f} MB")
print(f"   Tables: silver_transactions, gold_corridor_summary, gold_hourly_traffic,
print(f"\n📝 Cara import ke MySQL:")
print(f"   Option 1 (CLI):  mysql -u root -p < {SQL_PATH}")
print(f"   Option 2 (Workbench): File → Open SQL Script → Run")
print(f"   Option 3 (phpMyAdmin): Tab Import → Choose File → Go")
```

```
💾 EXPORT TO SQL FILE (.sql)
============================================================
  Generating SQL for silver_transactions...
  Generating SQL for gold_corridor_summary...
  Generating SQL for gold_hourly_traffic...
  Generating SQL for gold_demographics...
  Generating SQL for gold_daily_trend...

✅ Exported to SQL file: TransJakarta_Export.sql
   File size: 59.99 MB
   Tables: silver_transactions, gold_corridor_summary, gold_hourly_traffic, gold_dem
ographics, gold_daily_trend

📝 Cara import ke MySQL:
   Option 1 (CLI):  mysql -u root -p < TransJakarta_Export.sql
   Option 2 (Workbench): File → Open SQL Script → Run
   Option 3 (phpMyAdmin): Tab Import → Choose File → Go
```

In [25]:

```
# === EXPORT TO MySQL ===
print("💾 EXPORT TO MySQL")
print("=" * 60)

try:
    from sqlalchemy import create_engine

    # Konfigurasi MySQL — sesuaikan dengan setup lokal Anda
    MYSQL_USER = 'root'
    MYSQL_PASSWORD = ''  # Kosongkan jika tidak ada password
    MYSQL_HOST = 'localhost'
    MYSQL_PORT = '3306'
    MYSQL_DB = 'transjakarta_db'

    # Buat koneksi
    engine = create_engine(f'mysql+pymysql://{MYSQL_USER}:{MYSQL_PASSWORD}@{MYSQL_H

    # Test koneksi
    with engine.connect() as conn:
        print("✅ MySQL connection successful!")

    # Export tables
    silver_export.to_sql('silver_transactions', engine, if_exists='replace', index=
    print("✅ Exported Silver Layer to MySQL: silver_transactions")

    gold_corridor.to_sql('gold_corridor_summary', engine, if_exists='replace', inde
    print("✅ Exported Gold Corridor Summary to MySQL: gold_corridor_summary")

    gold_hourly.to_sql('gold_hourly_traffic', engine, if_exists='replace', index=Fa
    print("✅ Exported Gold Hourly Traffic to MySQL: gold_hourly_traffic")
```

```
    gold_demographics.to_sql('gold_demographics', engine, if_exists='replace', inde
    print("✅ Exported Gold Demographics to MySQL: gold_demographics")

    gold_daily.to_sql('gold_daily_trend', engine, if_exists='replace', index=False)
    print("✅ Exported Gold Daily Trend to MySQL: gold_daily_trend")

    print("\n✅ Semua data berhasil di-export ke MySQL!")

except ImportError:
    print("⚠️ pymysql tidak terinstall. Jalankan: pip install pymysql sqlalchemy")
    print("   Setelah install, jalankan ulang cell ini.")
except Exception as e:
    print(f"⚠️ MySQL connection failed: {e}")
    print("\n📝 Untuk menggunakan MySQL export:")
    print("   1. Pastikan MySQL Server sudah running")
    print("   2. Buat database: CREATE DATABASE transjakarta_db;")
    print("   3. Install library: pip install pymysql sqlalchemy")
    print("   4. Sesuaikan MYSQL_USER, MYSQL_PASSWORD, MYSQL_HOST di atas")
    print("   5. Jalankan ulang cell ini")
```

```
💾 EXPORT TO MySQL
============================================================
⚠️ MySQL connection failed: (pymysql.err.OperationalError) (2003, "Can't connect to
MySQL server on 'localhost' ([WinError 10061] No connection could be made because th
e target machine actively refused it)")
(Background on this error at: https://sqlalche.me/e/20/e3q8)

📝 Untuk menggunakan MySQL export:
   1. Pastikan MySQL Server sudah running
   2. Buat database: CREATE DATABASE transjakarta_db;
   3. Install library: pip install pymysql sqlalchemy
   4. Sesuaikan MYSQL_USER, MYSQL_PASSWORD, MYSQL_HOST di atas
   5. Jalankan ulang cell ini
```

✅ Summary

Proses yang telah dilakukan:

No	Proses	Status
1	Data Understanding & Eksplorasi	✅
2	Bronze Layer — Raw Data Ingestion	✅
3	Silver Layer — Data Cleansing	✅
4	Data Quality Check (Completeness, Consistency, Validity, Uniqueness)	✅
5	Privacy Policy — Data Masking (payCardID, payCardName)	✅
6	Gold Layer — Aggregated Tables (4 tabel)	✅
7	Metadata Strategy — Data Dictionary	✅
8	Data Analysis — 8 Visualisasi dengan Insight	✅

No	Proses	Status
9	Export ke Excel (.xlsx)	✅
10	Export ke MySQL	✅ / ⚠️

Key Insights:

1. **Pola Commuter**: Transaksi paling tinggi di jam sibuk pagi (06-09) dan sore (17-19)
2. **Weekday vs Weekend**: Transaksi weekday secara signifikan lebih tinggi dari weekend
3. **Demografi**: Mayoritas penumpang adalah usia produktif
4. **Gratis**: Terdapat proporsi signifikan transaksi Rp 0 (kemungkinan subsidi)
5. **Corridor**: Beberapa corridor mendominasi jumlah transaksi

---

Project Data Engineer | Dibimbing Dataset: Transjakarta - Public Transportation Transaction (Kaggle)